

FORMALIZATION OF LANGLANDS’S FIRST MAIN LEMMA FOR LOCAL EPSILON FACTORS

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ABSTRACT. We formalize in Lean 4 Langlands’s First Main Lemma for local epsilon factors over all nonarchimedean local fields. This equality of local constants in cyclic extensions of prime degree is one of the four Main Lemmas used in Langlands’s construction of epsilon factors of local Weil representations. The proof follows the author’s companion paper, which reconstructs and completes the local arguments of Dwork and Langlands, including the equal-characteristic case.

The First Main Lemma also follows from Deligne’s construction of local epsilon factors by global methods; here we formalize a purely local proof. The development includes the finite-sum and Haar-integral constructions of local constants, the required ramification and finite-field identities, and the comparison of stationary representatives under twisting and norm.

ChatGPT and Codex assisted with parts of the Lean development. The author used ChatGPT to check the manuscript for consistency and to suggest improvements to wording and organization of materials.

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1. INTRODUCTION

1.1. The First Main Lemma. Let E be a nonarchimedean local field with Weil group W_E . Langlands's local epsilon factors attached to representations of W_E are determined by their values on one-dimensional representations, multiplicativity, and compatibility with induction. Using Brauer induction, their construction is reduced to local epsilon factors of quasi-characters. One must then prove that the answer does not depend on the chosen Brauer decomposition. Langlands reduced this to four basic identities, which he called the Main Lemmas. In this paper we formalize the *First Main Lemma*; see [3]¹.

To a quasi-character $\chi_E : E^\times \rightarrow \mathbf{C}^\times$ and a nontrivial additive character $\psi_E : E \rightarrow \mathbf{C}^\times$, Langlands associates a nonzero complex number $\Delta(\chi_E, \psi_E)$, which is essentially a normalized Gauss sum. He also defines the epsilon factor

$$\varepsilon(s, \chi_E, \psi_E) := \Delta\left(|\cdot|_E^{s-\frac{1}{2}} \chi_E, \psi_E\right),$$

where $|x|_E = q_E^{-v_E(x)}$, with q_E the cardinality of the residue field of E . Thus

$$\Delta(\chi_E, \psi_E) = \varepsilon\left(\frac{1}{2}, \chi_E, \psi_E\right).$$

On the other hand, denoting by m and n the conductors of χ_E and ψ_E , respectively, we have

$$\varepsilon(s, \chi_E, \psi_E) = q_E^{-(m+n)(s-\frac{1}{2})} \Delta(\chi_E, \psi_E).$$

Let K/F be a cyclic extension of prime degree ℓ of nonarchimedean local fields. For a quasi-character χ_F and a nontrivial additive character ψ_F , put

$$\chi_K = \chi_F \circ N_{K/F}, \quad \psi_K = \psi_F \circ \mathrm{Tr}_{K/F}.$$

Let

$$S(K/F) = \{\mu \in \mathrm{Hom}_{\mathrm{cont}}(F^\times, \mathbf{C}^\times) : \mu \circ N_{K/F} = 1\}$$

be the group of norm characters.

Langlands's First Main Lemma is the equality

$$(1) \quad \Delta(\chi_K, \psi_K) \prod_{\mu \in S(K/F)} \Delta(\mu, \psi_F) = \prod_{\mu \in S(K/F)} \Delta(\mu \chi_F, \psi_F).$$

The local proof goes back to Dwork and Langlands. Dwork's 1954 thesis contains the main p -adic calculations [2], while Langlands stated the First Main Lemma and organized the proof by ramification theory and stationary phase [3]. After Deligne [1] found a shorter proof using global functional equations, Langlands stopped preparing a complete version of the local argument.

Dwork used different normalizations and worked in mixed characteristic. Langlands's surviving proof of the First Main Lemma ends before completing the wild endpoint cases, the non-stable wild odd-prime case, and the wild quadratic case; see [Section 2.5](#) for the complete case

¹Chapter 10, entitled "The First Main Lemma," is absent from the surviving manuscript. The identity is nevertheless reproduced in [3, (12.1), p. 144], where it is attributed to Lemma 10.1.

division. In the companion paper [4], we put these arguments in one notation, complete the remaining calculations, and include the equal-characteristic case. The companion paper gives the mathematical proof; here we describe its formalization in Lean 4 [6], using Mathlib [7].

1.2. The theorem proved in Lean. The proposition expressing the First Main Lemma is `LanglandsFirstMainLemma.FirstMainIdentity`, and the main theorem proving it is `LanglandsFirstMainLemma.firstMainLemma`.

Theorem 1.1 (Formalized First Main Lemma). *Let K/F be a cyclic extension of prime degree between nonarchimedean local fields. For every continuous quasi-character χ_F of $F^\times$ and every nontrivial continuous additive character ψ_F of F , the equality (1) holds.*

The statement has no characteristic-zero hypothesis. Mixed and equal characteristic are treated separately in the final correction calculations. In order to form the product in (1) it is necessary to show that $S(K/F)$ is finite. The formalization proves this via basic ramification theory.

We compared the statement and all the definitions which occur in it with the companion paper. They agree.

1.3. The formal proof. The formal proof follows the companion mathematical proof. We explain below how its choices, conductor conditions, and finite-sum calculations are represented in Lean.

First, both proofs establish nonvanishing of local constants through finite Gauss sums and finite Fourier duality before cancellation. This differs from Langlands's presentation, which starts from the Haar-integral definition and refers to a later part of the argument for nonvanishing.

Second, the mathematical proof repeatedly makes choices of representatives and auxiliary elements. In Lean, the relation between the old and new choices has to be recorded, and the final expression has to be proved independent of them.

Third, several parts of the argument depend on parity, conductor, or ramification conditions. The formalization follows the same case division as the companion proof and records the hypotheses of each case.

Some specific examples of these issues are discussed later, after the corresponding mathematical objects have been introduced.

1.4. Plan of the paper. Section 2 recalls the local constant, the First Main Lemma, and the division of the proof into cases. In Section 3 we give the exact Lean statement and explain its assumptions. The construction of the local constant and the formal version of Lamprecht's formula are discussed in Section 4. The ramification and finite-field results used in the proof are described in Section 5, and the individual cases in Section 6. We record reusable results in Section 7 and describe the released artifact in Section 8.

2. MATHEMATICAL BACKGROUND AND PROOF

We use the notation of the companion paper [4].

2.1. Local fields and conductors. Let E be a nonarchimedean local field. Write

$$\mathcal{O}_E = \{x \in E : v_E(x) \geq 0\}, \quad \mathfrak{p}_E = \{x \in E : v_E(x) \geq 1\},$$

let $k_E = \mathcal{O}_E/\mathfrak{p}_E$ be the residue field, whose cardinality is denoted q_E , and normalize $v_E(E^\times) = \mathbf{Z}$. Put

$$U_E^0 = \mathcal{O}_E^\times, \quad U_E^r = 1 + \mathfrak{p}_E^r \quad (r \geq 1).$$

The quasi-characters are continuous homomorphisms $E^\times \rightarrow \mathbf{C}^\times$, and the additive characters are continuous homomorphisms from the additive group of E to $\mathbf{C}^\times$.

Write $p = \text{char } k_E$. An integer in E means its image under $\mathbf{Z} \rightarrow E$. In particular, $p = 0$ when $\text{char } E = p$. This convention will be important when the same formula is used in both mixed and equal characteristic.

For a quasi-character $\theta : E^\times \rightarrow \mathbf{C}^\times$, the multiplicative conductor $m_E(\theta)$ is the least $m \geq 0$ such that the quasi-character θ is trivial on U_E^m . Thus $m_E(\theta) = 0$ exactly when θ is unramified.

For a nontrivial additive character $\psi : E \rightarrow \mathbf{C}^\times$, the integer $n_E(\psi)$ is defined by requiring $\mathfrak{p}_E^{-n_E(\psi)}$ to be the largest fractional ideal on which ψ is trivial.

These conventions are formalized in Section 4.1 and used throughout the case analysis in Section 6.

2.2. Langlands's local constant. For $z \in \mathbf{C}$, set

$$\text{ph}(z) = \begin{cases} z/|z|, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

Let $m = m_E(\theta)$ and $n = n_E(\psi)$, and choose $\gamma \in E^\times$ with

$$(2) \quad \gamma \mathcal{O}_E = \mathfrak{p}_E^{m+n}.$$

Such an element γ is called admissible. For a positive Haar measure du on U_E^0 , Langlands defines

$$(3) \quad \Delta(\theta, \psi) = \theta(\gamma) \text{ph} \left(\int_{U_E^0} \psi(u/\gamma) \theta(u)^{-1} du \right).$$

The value does not depend on the positive normalization of du or on the choice of the admissible element γ .

There is also a finite form of the same constant. Put $Q_m = U_E^0/U_E^m$ and define

$$(4) \quad G(\theta, \psi; \gamma) = \sum_{\bar{u} \in Q_m} \psi(u/\gamma) \theta(u)^{-1}.$$

The summand depends only on the class of u . The quotient Q_m is naturally the unit group of the finite ring $\mathcal{O}_E/\mathfrak{p}_E^m$. Thus (4) is a finite Gauss sum. When $m = 1$ it is the usual Gauss sum over $k_E^\times$.

The finite sum is nonzero, and the local constant has the finite formula

$$(5) \quad \Delta(\theta, \psi) = \theta(\gamma) \text{ph}(G(\theta, \psi; \gamma)).$$

This finite realization and its nonvanishing are [4, Proposition 2.4].

The realizations (3) and (5) are compared formally in Section 4.3 and used in every case of Section 6.

2.3. The First Main Lemma and the first reduction. Recall the setup from Section 1. The First Main Lemma is (1).

It is proved in [4, Corollary 3.12] that

$$(6) \quad |S(K/F)| = |F^\times/N_{K/F}(K^\times)| = [K : F].$$

Local class field theory is not used for this step or elsewhere in the proof.

Define the error quotient

$$(7) \quad \mathcal{E}_{K/F}(\chi_F) = \frac{\Delta(\chi_K, \psi_K) \prod_{\mu \in S(K/F)} \Delta(\mu, \psi_F)}{\prod_{\mu \in S(K/F)} \Delta(\mu \chi_F, \psi_F)}.$$

The nonvanishing stated in Section 2.2 makes this quotient well defined. The First Main Lemma is equivalent to $\mathcal{E}_{K/F}(\chi_F) = 1$.

If $\mu \in S(K/F)$, then multiplication by μ permutes $S(K/F)$. Therefore

$$(8) \quad \mathcal{E}_{K/F}(\mu \chi_F) = \mathcal{E}_{K/F}(\chi_F).$$

We may consequently replace χ_F by a character of minimal conductor in its $S(K/F)$ -orbit.

The case proofs in Section 6 use this reduction.

2.4. Stationary classes and Lamprecht's formula. The stationary class plays a central role in the proof of the First Main Lemma. The companion paper constructs the stationary class, proves Lamprecht's formula, and proves independence of the chosen representative in [4, Lemma 4.2, Theorem 4.5, and Lemma 4.6]. The formal stationary class and Lamprecht's formulas are given in Sections 4.4 and 4.5.

The stationary class of a character is used when its multiplicative conductor is greater than one. Individual local constants of conductor zero or one are evaluated directly. In the wild conductor-one case, however, the norm characters have higher conductor, so their stationary classes are still needed.

Assume $m_E(\theta) > 1$. On a sufficiently deep unit group, the multiplicative character θ can be written in terms of the additive character ψ . More precisely, consider an integer M , an element $\Gamma \in E^\times$ with $\Gamma \mathcal{O}_E = \mathfrak{p}_E^{M+n_E(\psi)}$, and an integer $\lceil m_E(\theta)/2 \rceil \leq r \leq m_E(\theta) - 1$. There is a unique class

$$(9) \quad \text{St}_{\Gamma,r}(\theta) \in \mathfrak{p}_E^{M-m_E(\theta)} / \mathfrak{p}_E^{M-r}$$

with the following property: if $\beta = \beta(\theta)$ represents this class, then

$$(10) \quad \theta(1+x) = \psi(\beta x / \Gamma) \quad (x \in \mathfrak{p}_E^r).$$

The element β is not unique, but its class in (9) is.

Write

$$m_E(\theta) = 2d + \varepsilon > 1, \quad \varepsilon \in \{0, 1\}.$$

Apply this to the special case that $\Gamma = \gamma$ is an admissible element for (θ, ψ) and

$$M = m_E(\theta), \quad r = d + \varepsilon.$$

The stationary class then lies in $\mathcal{O}_E / \mathfrak{p}_E^d$, and every representative β is a unit.

If the conductor is even, Lamprecht's formula gives

$$\Delta(\theta, \psi) = \theta(\gamma) \psi(\beta/\gamma) \theta(\beta)^{-1}.$$

If the conductor is odd, the same expression is multiplied by the phase of a finite sum over the residue field. Choose a critical element $\delta \in \mathfrak{p}_E^d \setminus \mathfrak{p}_E^{d+1}$ and define the *Lamprecht critical function* by

$$(11) \quad \phi_{\theta,\beta,\delta}(\bar{x}) = \phi_{\theta,\beta}(\bar{x}) = \psi(\delta \beta \bar{x} / \gamma) \theta(1 + \delta \bar{x})^{-1}, \quad \bar{x} \in k_E,$$

where $\tilde{x} \in \mathcal{O}_E$ is any lift of $\bar{x}$. We suppress the critical element δ from $\phi_{\theta,\beta}$ to lighten the notation; the function itself may depend on δ , but the phase of its sum over k_E does not.

The critical function is well defined and satisfies

$$(12) \quad \phi_{\theta,\beta}(x+y) = \phi_{\theta,\beta}(x) \phi_{\theta,\beta}(y) \eta(xy), \quad \eta(\bar{z}) = \psi\left(\frac{\beta \delta^2 \bar{z}}{\gamma}\right).$$

The character η is nontrivial, so $\phi_{\theta,\beta}$ is a Hasse function in the sense of Section 5.3.2. These properties are part of [4, Theorem 4.5]. Write

$$\mathcal{A}_E(\theta; \beta, \gamma) = \theta(\gamma) \psi(\beta/\gamma) \theta(\beta)^{-1}, \quad \mathcal{H}_E(\theta; \beta, \gamma) = \begin{cases} 1, & \varepsilon = 0, \\ \text{ph} \left(\sum_{\bar{x} \in k_E} \phi_{\theta,\beta}(\bar{x}) \right), & \varepsilon = 1. \end{cases}$$

Then Lamprecht's formula is

$$(13) \quad \Delta(\theta, \psi) = \mathcal{A}_E(\theta; \beta, \gamma) \mathcal{H}_E(\theta; \beta, \gamma).$$

The first factor in (13) is expressed entirely in the local field. The second factor is either 1 or the phase of a finite sum over the residue field. A change of the representative β may change these two factors separately, but it does not change their product.

To prove the First Main Lemma, one applies (13) to each local constant of conductor greater than one among

$$\Delta(\chi_K, \psi_K), \quad \Delta(\mu, \psi_F), \quad \Delta(\mu \chi_F, \psi_F).$$

The proof then has two parts.

First, one compares the products of the local-field factors $\mathcal{A}_E(\theta; \beta, \gamma)$. If, on the same unit layer,

$$\mu(1+x) = \psi_F(ax/\gamma), \quad \chi_F(1+x) = \psi_F(bx/\gamma),$$

then

$$(\mu\chi_F)(1+x) = \psi_F((a+b)x/\gamma).$$

Thus $a+b$ gives the character identity for $\mu\chi_F$ on the same unit group. If its conductor drops, the stationary class is formed at the new conductor when this is greater than one; otherwise we use the direct formulas for conductor zero or one.

Suppose that χ_F and $\chi_K = \chi_F \circ N_{K/F}$ both have conductor greater than one. Choose admissible elements γ_F, γ_K for (χ_F, ψ_F) and (χ_K, ψ_K) , respectively, and put

$$P_{K/F}(x) = N_{K/F}(1+x) - 1.$$

Choose a stationary depth r_K for χ_K such that $P_{K/F}(x)$ lies in the linearization domain of χ_F for $x \in \mathfrak{p}_K^{r_K}$. For a corresponding representative β_F , we have

$$\chi_K(1+x) = \psi_F\left(\frac{\beta_F P_{K/F}(x)}{\gamma_F}\right).$$

The stationary class of χ_K at depth r_K is therefore the unique class whose representatives β_K satisfy

$$(14) \quad \psi_F\left(\frac{\beta_F P_{K/F}(x)}{\gamma_F}\right) = \psi_K\left(\frac{\beta_K x}{\gamma_K}\right), \quad x \in \mathfrak{p}_K^{r_K}.$$

Computing this class in each ramification case gives the compatible representatives needed to compare the factors $\mathcal{A}_F$ and $\mathcal{A}_K$.

Second, one compares the residue-field factors $\mathcal{H}_E(\theta; \beta, \gamma)$. These are finite-field calculations: Hasse–Davenport identities, Hasse–function lifting, quadratic Gauss sums, or the characteristic-two Artin–Schreier identity; the corresponding results are stated in Sections 5.3.1–5.3.6.

The case division in Section 2.5 below records the different ways in which the stationary classes are related under twist and norm, and the finite-field identities needed after that comparison has been made.

2.5. The cases of the proof. Let K/F be a finite Galois extension of nonarchimedean local fields, and write $G = \text{Gal}(K/F)$. For $i \geq 0$, its lower ramification groups are

$$G_i = \{\sigma \in G : \sigma(x) \equiv x \pmod{\mathfrak{p}_K^{i+1}} \text{ for every } x \in \mathcal{O}_K\}.$$

They form a decreasing sequence of subgroups, with $G_i = 1$ for all sufficiently large i .

Now suppose that K/F is cyclic of prime degree ℓ , and put $p = \text{char } k_F$. If K/F is unramified, then $G_i = 1$ for every $i \geq 0$. Otherwise, K/F is totally ramified, so $G_0 = G$. Since a group of prime order has no proper nontrivial subgroup, there is a unique integer $t \geq 0$ such that

$$G_0 = \cdots = G_t = G, \quad G_{t+1} = 1.$$

This integer t is the lower ramification break. In the tamely ramified case, $\ell \neq p$ and $t = 0$; in the wildly ramified case, $\ell = p$ and $t \geq 1$. In the ramified cases, put

$$T = t + 1.$$

By the twist invariance (8), we may choose χ_F of minimal conductor in its $S(K/F)$ -orbit. Write

$$m = m_F(\chi_F) = 2d + \varepsilon, \quad d = \lfloor m/2 \rfloor, \quad \varepsilon \in \{0, 1\}.$$

The proof is divided into the following cases, as in [4, Sections 7–10].

Extension and conductor	Reduction and later references
K/F unramified	For $m = 0$, use the direct formula for an unramified character. For $m = 1$, use the Hasse–Davenport lifting identity. For $m > 1$, compatible stationary representatives reduce the odd-conductor residue-field sum to the Hasse-function lifting identity; when m is even, there is no residue-field sum. See (38), (44), and Sections 5.3.1, 5.3.2, and 6.1.
K/F tamely ramified, ℓ odd	For $m = 0$, use the direct local-field formulas. For $m = 1$, use the Hasse–Davenport product identity. For $m > 1$, every nontrivial $\mu \in S(K/F)$ has conductor one and does not change the stationary class of χ_F ; the remaining odd-conductor residue-field phases satisfy the ramified Hasse-phase identity. See (39), (49), and Sections 5.3.1, 5.3.4, and 6.2.
K/F tamely ramified, $\ell = 2$	For $m = 0$ and $m = 1$, the proof is the same as above. For $m > 1$, the same stationary representative may be used for χ_F , for its twist by the nontrivial member of $S(K/F)$, and for χ_K ; the remaining residue-field identity is evaluated by quadratic Gauss sums. See (48), and Sections 5.3.3 and 6.2.
K/F wild, $m \leq 1$	The conductor-zero case is again direct. In conductor one, Frobenius identifies the Gauss sums for χ_F and χ_K , while (32) determines the remaining factors involving the nontrivial norm characters. The resulting comparison is completed by (68). See Sections 5.2 and 6.3.
K/F wild, $p > 2$, $m \geq 2T$	This is the stable range. Every nontrivial $\mu \in S(K/F)$ has conductor $T \leq d$, so multiplication by μ does not change the stationary class of χ_F . The First Main Lemma reduces to (69). The local-field factors are compared under norm, and the odd-conductor residue-field phases are compared by (49). See Sections 5.3.4 and 6.4.
K/F wild, $p > 2$, $1 < m < 2T$	Here $T > d$, so multiplication by norm characters can change the stationary class. After choosing a generator τ of $S(K/F)$, the stationary classes of τ^j , $\tau^j \chi_F$, and χ_K are computed explicitly. The residue-field phases are evaluated by the quadratic-phase formula and, when $m = T$, by the affine-pencil identity (51); the remaining local-field term is evaluated by (76). See Sections 5.2, 5.3.3, 5.3.5, and 6.5.
K/F wild, $p = 2$, $m > 1$	Write $S(K/F) = \{1, \tau\}$. The stationary classes of τ , χ_F , $\tau \chi_F$, and χ_K are compared explicitly. Their residue-field phases are expressed through one characteristic-two quadratic refinement. In the exceptional boundary case, the phases and remaining character values cancel by (58). See Sections 5.3.6 and 6.6.

The ramification, norm, trace, and finite-field statements used in these comparisons are valid in both mixed and equal characteristic. The characteristic distinction appears only in the last correction terms of the wild cases. In equal characteristic those terms vanish in the field. In mixed characteristic their valuations make the additive contribution trivial, except at the wild quadratic odd boundary. There the additive contribution cancels together with the critical-function values and residue-field phases.

The theorem `firstMainLemma_nonarchimedean` combines these cases by invoking the corresponding result.

3. THE EXACT LEAN THEOREM

3.1. The formal statement of the First Main Lemma. We now match the terms of the First Main Lemma (1) with their Lean definitions.

The type `ContinuousQuasiCharF` consists of continuous homomorphisms

$$\chi_F : F^\times \longrightarrow \mathbf{C}^\times,$$

and `ContinuousAddCharF` consists of continuous additive characters

$$\psi_F : F \longrightarrow \mathbf{C}^\times.$$

The two pullbacks in the theorem are represented by

$$\text{normQuasiChar } F \ K \ \chi_F = \chi_F \circ N_{K/F}$$

and

$$\text{tracePullbackAddChar } F \ K \ \psi_F = \psi_F \circ \text{Tr}_{K/F}.$$

The type `NormCharacterFK` represents

$$S(K/F) = \{\mu \in \text{Hom}_{\text{cont}}(F^\times, \mathbf{C}^\times) : \mu \circ N_{K/F} = 1\}.$$

The type `LocalConstantFunctionE` consists of functions assigning a complex number to each pair (χ, ψ) of a continuous quasi-character χ of $E^\times$ and a continuous additive character ψ of E . The arguments `DeltaF` and `DeltaK` have types `LocalConstantFunctionF` and `LocalConstantFunctionK`, respectively. In the exported theorem below, `DeltaF` and `DeltaK` are instantiated by the formalized local constants `localConstantF` and `localConstantK`.

The functions `firstMainLeftSide` and `firstMainRightSide` represent the two sides of (1). Their equality is the proposition:

```
def FirstMainIdentity
  (ΔF : LocalConstantFunction F) (ΔK : LocalConstantFunction K)
  (χF : ContinuousQuasiChar F) (ψF : ContinuousAddChar F) : Prop :=
  firstMainLeftSide F K ΔF ΔK χF ψF =
  firstMainRightSide F K ΔF ΔK χF ψF
```

The definition `FirstMainIdentity` is made under the assumption

$$[\text{Finite } (\text{NormCharacter } F \ K)],$$

since its two sides contain products over $S(K/F)$. In the exported theorem, the finiteness of $S(K/F)$ is not assumed separately. The declaration `primeCyclicNormCharacter_finite` proves the finiteness of `NormCharacterFK` for a cyclic extension of prime degree, as required to form the products in the First Main Lemma.

3.2. The exported theorem. The final declaration is `firstMainLemma`:

```
theorem firstMainLemma
  (F K : Type)
  [Field F] [ValuativeRel F] [TopologicalSpace F]
  [IsNonarchimedeanLocalField F]
  [Field K] [ValuativeRel K] [TopologicalSpace K]
  [IsNonarchimedeanLocalField K]
  [Algebra F K] [ValuativeExtension F K]
  [Module.Free F K] [Module.Finite F K]
  (hcyclic : CyclicPrimeExtension F K) :
  let I : PrimeCyclicExtension F K :=
    PrimeCyclicExtension.ofCyclicPrimeExtension F K hcyclic
  let I : Finite (NormCharacter F K) :=
    primeCyclicNormCharacter_finite F K
  ∀ (χF : ContinuousQuasiChar F) (ψF : ContinuousAddChar F),
  ψF ≠ 1 →
    FirstMainIdentity F K (localConstant F) (localConstant K) χF ψF
```

Here `hcyclic` is the hypothesis that K/F is cyclic of prime degree. The first `letI` line makes this same hypothesis available to the ramification lemmas as `PrimeCyclicExtensionFK`. It does not add another assumption.

The second `letI` line uses the finiteness theorem cited in Section 3.1 for `NormCharacterFK`, which represents $S(K/F)$; the finiteness of the latter is proved from `hcyclic` and is not an extra hypothesis.

The theorem then quantifies over a continuous quasi-character χ_F and a continuous additive character ψ_F , with the assumption that ψ_F is nontrivial. Its conclusion is

`FirstMainIdentity F K (localConstant F) (localConstant K) chiF psiF.`

By the definitions in Section 3.1, this is exactly the conclusion (1) in the First Main Lemma.

3.3. The assumptions. The assumptions before `hcyclic` describe the local fields and the extension K/F .

Lean assumption	Mathematical meaning
<code>FieldF</code> , <code>ValuativeRelF</code> , <code>TopologicalSpaceF</code> , <code>IsNonarchimedeanLocalFieldF</code>	F is a nonarchimedean local field with its valuation and topology.
The corresponding assumptions on K	K is a nonarchimedean local field.
<code>AlgebraFK</code>	A specified embedding $F \rightarrow K$.
<code>ValuativeExtensionFK</code>	The valuations on F and K are compatible with this embedding.
<code>Module.FreeFK</code> , <code>Module.FiniteFK</code>	K is finite free as an F -vector space. This is the form used by Mathlib for a finite extension.
<code>CyclicPrimeExtensionFK</code>	K/F is Galois, its Galois group is cyclic, and $[K : F]$ is prime.

The declaration is written with `(FK:Type)`, so it lies in universe zero. This restricts only the universe generality of the Lean statement and has no effect on the local fields considered here.

3.4. The trivial additive character and the phase at zero. Lean defines `localConstantE` for every continuous additive character, so that it is an ordinary total function. Its value at the trivial additive character is set to zero. This convention does not enter the First Main Lemma. The exported theorem assumes that ψ_F is nontrivial, and the proof also establishes the nontriviality of its trace pullback to K .

The phase function is treated similarly:

`phase 0 = 1.`

Before the proof uses a phase arising from a Gauss sum or a Haar integral, that sum or integral is proved to be nonzero. Thus the value assigned at zero is not used in the proof of the theorem.

4. LAMPRECHT'S FORMULA IN LEAN

This section formalizes the local constant and stationary classes recalled in Sections 2.2 and 2.4. These declarations are used in the case proofs of Section 6.

4.1. Exact conductors and admissible elements. Lean represents an exact conductor by a predicate involving a character and an integer. The statement

`IsMultiplicativeConductor F chi m`

means that $m = m_F(\chi)$, and

`IsAdditiveConductor F psi n`

means that $n = n_F(\psi)$. The structure `LocalQuasiCharDataF` records a quasi-character together with its exact multiplicative conductor, while `LocalAddCharDataF` records a nontrivial additive character together with its exact additive conductor.

The existence and uniqueness theorems for conductors show that every continuous quasi-character χ_F , and every nontrivial continuous additive character ψ_F , determine canonical data of these two types. Accordingly, the proof of `firstMainLemma` replaces `chiF` and `psiF` by

`canonicalLocalQuasiCharData F chiF`

and

`canonicalLocalAddCharData F psiF hpsiF,`

where `hpsiF` is the hypothesis $\psi_F \neq 1$.

For

`chi : LocalQuasiCharData F,    psi : LocalAddCharData F,`

the type `AdmissibleGammaFchipsi`, defined by `AdmissibleGamma`, consists of elements $\gamma \in F^\times$ satisfying

$$v_F(\gamma) = m_F(\chi) + n_F(\psi).$$

This is the Lean form of (2), the condition defining admissible elements.

4.2. The finite Gauss sum. Let $m = m_F(\chi)$ and let γ be admissible. For $u \in U_F^0$, consider

$$\psi(u/\gamma)\chi(u)^{-1}.$$

The theorem `finiteGaussSummandRepresentative_eq_of_congruent` states that this value is unchanged when u is replaced by another unit with the same image in U_F^0/U_F^m . The expression therefore defines a function on the finite quotient U_F^0/U_F^m .

Lean defines its sum by the declaration `finiteGaussSum`:

$$(15) \quad \text{finiteGaussSum chi psi gamma} = \sum_{\bar{u} \in U_F^0/U_F^m} \psi(u/\gamma)\chi(u)^{-1},$$

and then defines `deltaFinite`:

$$(16) \quad \text{deltaFinite chi psi gamma} = \chi(\gamma) \text{ph}(\text{finiteGaussSum chi psi gamma}).$$

At this stage, `deltaFinite` denotes the finite expression constructed from the quotient sum. Its equality with Langlands's Haar-integral definition of $\Delta(\chi, \psi)$ is proved in Section 4.3.

The theorem `finiteGaussSum_ne_zero` proves

$$(17) \quad \text{finiteGaussSum chi psi gamma} \neq 0.$$

For positive conductor, its Lean proof extends the function on the unit classes by zero to the finite additive group $\mathcal{O}_F/\mathfrak{p}_F^m$ and applies finite Fourier inversion. The conductor-zero case is treated separately.

Finally, let γ' be another admissible element. The theorem `deltaFinite_gamma_independent` is precisely the equality

$$\text{deltaFinite chi psi gamma} = \text{deltaFinite chi psi gamma'}.$$

Thus the finite expression is independent of the admissible element used to define it.

4.3. Comparison with the Haar integral. Let $m = m_F(\chi)$, let γ be admissible, and let μ_H be a positive Haar measure on the compact multiplicative group U_F^0 .

The Lean definition `unitHaarIntegrand` is the function

$$u \mapsto \psi(u/\gamma)\chi(u)^{-1},$$

and `unitGaussIntegral` is the integral

$$(18) \quad I_{\mu_H}(\chi, \psi; \gamma) = \int_{U_F^0} \psi(u/\gamma)\chi(u)^{-1} d\mu_H(u).$$

Lean then defines

$$(19) \quad \text{deltaIntegral } \mu_H \chi \psi \gamma = \chi(\gamma) \text{ ph}(I_{\mu_H}(\chi, \psi; \gamma)).$$

Thus `deltaIntegral` is Langlands's integral expression with the Haar measure and the admissible denominator still shown as arguments.

The comparison with the finite sum (15) comes from the coset decomposition

$$(20) \quad U_F^0 = \coprod_{\bar{a} \in U_F^0/U_F^m} aU_F^m.$$

The integrand in (18) is constant on each coset. Indeed, if $v \in U_F^m$, then

$$\frac{\psi(av/\gamma)\chi(av)^{-1}}{\psi(a/\gamma)\chi(a)^{-1}} = \psi\left(\frac{a(v-1)}{\gamma}\right)\chi(v)^{-1} = 1.$$

By invariance of Haar measure, every coset in (20) has the same measure. Put

$$c_{\mu_H, m} = \mu_H(U_F^m).$$

It is a positive real number; see `integralFinitePositiveScalar_pos`. Integrating the constant value on each coset gives

$$(21) \quad I_{\mu_H}(\chi, \psi; \gamma) = c_{\mu_H, m} G(\chi, \psi; \gamma).$$

This is the theorem `unitGaussIntegral_eq_positiveScalar_mul_finiteGaussSum`, where the sum on the right is `finiteGaussSum` from (15).

Since $c_{\mu_H, m} > 0$ and the finite Gauss sum is nonzero by (17), the integral is nonzero and has the same phase as the sum by (21). These conclusions are proved by `unitGaussIntegral_ne_zero_of_finiteGaussSum` and `deltaIntegral_eq_deltaFinite`:

$$(22) \quad \text{deltaIntegral } \mu_H \chi \psi \gamma = \text{deltaFinite } \chi \psi \gamma.$$

The integral depends on the Haar measure, but its phase does not. The independence of Δ from the measure and from the admissible element is proved by `deltaIntegral_measure_independent` and `deltaIntegral_gamma_independent_of_ne_zero`.

The exported function `localConstantF` omits these auxiliary choices. For a nontrivial additive character it agrees with both realizations (16) and (19), by `localConstant_eq_deltaFinite` and `localConstant_eq_deltaIntegral`. Thus it realizes the Δ of (3).

4.4. The stationary class. Section 2.4 defined the stationary class in (9)

$$\text{St}_{\Gamma, r}(\theta) \in \mathfrak{p}_E^{M-m_E(\theta)} / \mathfrak{p}_E^{M-r}$$

by the identity (10):

$$\theta(1+x) = \psi(\beta x/\Gamma) \quad (x \in \mathfrak{p}_E^r),$$

where β is any representative of the class. The quotient class itself is `stationaryNumeratorClass`.

The theorem `stationaryNumeratorClass_existsUnique` is the formal counterpart of the existence and uniqueness statement in (9)–(10). For a chosen representative β , `stationaryNumeratorClass_representative_ord` proves

$$v_E(\beta) = M - m_E(\theta).$$

In the specialization used in Lamprecht's formula, $M = m_E(\theta)$, so this last equality says that every representative is a unit.

4.5. Lamprecht's formula. Let $m_E(\theta) = 2d + \varepsilon > 1$, with $\varepsilon \in \{0, 1\}$, let γ be admissible, and choose a representative $\beta \in \mathcal{O}_E^\times$ of $\text{St}_{\gamma, d+\varepsilon}(\theta)$.

When $\varepsilon = 0$, Lamprecht's formula is

$$(23) \quad \Delta(\theta, \psi) = \theta(\gamma)\psi(\beta/\gamma)\theta(\beta)^{-1}.$$

The Lean theorem `lamprechtEven` proves the corresponding equality for `deltaFinite`. By `deltaIntegral_eq_deltaFinite` (22), proved in Section 4.3, this is also the Haar-integral formula (23).

When $\varepsilon = 1$, recall the Lamprecht critical function (11) $\phi_{\theta, \beta}$ on the residue field k_E . Lean represents this function by `lamprechtHasseFunction`; the theorem `lamprechtOdd_hasseSum_ne_zero` proves the nonvanishing of its sum. The odd-conductor formula is

$$(24) \quad \Delta(\theta, \psi) = \theta(\gamma)\psi(\beta/\gamma)\theta(\beta)^{-1} \text{ph} \left(\sum_{x \in k_E} \phi_{\theta, \beta}(x) \right).$$

Its Lean counterpart is `lamprechtOdd`, again stated first for `deltaFinite`.

The right side of (24) contains two parts which both depend on the chosen representative β . The theorem `lamprechtOdd_complete_representative_independent` proves exactly that the complete right side is unchanged when β is replaced by another representative of the same stationary class.

Equations (23) and (24) are used in the higher-conductor case proofs in Sections 6.1, 6.2, 6.4, 6.5, and 6.6.

5. RAMIFICATION AND FINITE-FIELD IDENTITIES

The case proofs use norm and trace calculations in K/F to compare the local constants. For characters of odd conductor greater than one, Lamprecht's formula also leaves a residue-field sum. The finite-field identities below evaluate these sums and the Gauss sums in the conductor-one cases. Our main sources are the norm and trace results of [4, Section 3] and the finite-field identities of [4, Section 5].

5.1. The norm filtration. Let K/F be ramified cyclic of prime degree ℓ , and let t be its lower ramification break, as in Section 2.5. Put

$$T = t + 1, \quad D = (\ell - 1)T,$$

and, for an integer $r \geq 0$, put

$$\Psi(r) = \begin{cases} r, & 0 \leq r \leq t, \\ t + \ell(r - t), & r > t. \end{cases}$$

Thus $\Psi(r)$ is the depth in K corresponding to the depth r in F .

The trace of each fractional ideal is

$$(25) \quad \text{Tr}_{K/F}(\mathfrak{p}_K^s) = \mathfrak{p}_F^{\lfloor (s+D)/\ell \rfloor} \quad (s \in \mathbf{Z}).$$

In particular,

$$(26) \quad n_K(\psi_K) = \ell n_F(\psi_F) + D.$$

For every $r \geq 0$, the norm induces a map

$$N_r : U_K^{\Psi(r)} / U_K^{\Psi(r)+1} \longrightarrow U_F^r / U_F^{r+1}.$$

This map is an isomorphism for $r < t$ and for $r > t$. At $r = t$, its kernel and cokernel both have order ℓ .

The declaration `cyclicPrimeNormFiltration` proves the trace-ideal equality and the norm statements above.

Let $m = m_F(\chi_F)$. For $m > T$, the norm filtration gives

$$(27) \quad m_K(\chi_F \circ N_{K/F}) = \Psi(m - 1) + 1 = \ell m - D.$$

If χ_F has minimal conductor in its $S(K/F)$ -orbit, then

$$(28) \quad m_K(\chi_F \circ N_{K/F}) = m \quad (0 \leq m \leq T).$$

At $m = T$, minimality prevents the conductor from dropping. These statements follow from [4, Proposition 3.10].

For comparison, if K/F is unramified of degree f , then trace and norm preserve the filtration depths:

$$\mathrm{Tr}_{K/F}(\mathfrak{p}_K^s) = \mathfrak{p}_F^s, \quad N_{K/F}(U_K^r) = U_F^r.$$

Hence

$$(29) \quad n_K(\psi_F \circ \mathrm{Tr}_{K/F}) = n_F(\psi_F), \quad m_K(\chi_F \circ N_{K/F}) = m_F(\chi_F).$$

These are formalized by `unramified_additiveConductor_compTrace` and `unramified_multiplicativeConductor_compNorm`. They are used in Section 6.1; the ramified formulas above are used in Sections 5.2, 6.2, 6.3, 6.4, 6.5, and 6.6.

5.2. The norm at the wild break. Let K/F be wildly ramified and cyclic of prime degree p . Then p is the residue characteristic, and the residue fields of F and K are canonically the same finite field; write this field as k . Let $t > 0$ be the unique lower ramification break. Choose a uniformizer ϖ_K of K , put $\varpi_F = N_{K/F}(\varpi_K)$.

Both sides of the critical norm map

$$N_t : U_K^t/U_K^{t+1} \longrightarrow U_F^t/U_F^{t+1}$$

are naturally identified with the additive group of k . If $z \in k$ and $\tilde{z} \in \mathcal{O}_F$ is any lift, the source coordinate is

$$k \longrightarrow U_K^t/U_K^{t+1}, \quad z \longmapsto [1 + \tilde{z}\varpi_K^t],$$

and the target coordinate is defined in the same way using ϖ_F . These coordinates are well defined. In these coordinates the critical norm map is the function $P : k \rightarrow k$ determined by

$$(30) \quad P(z) = \frac{\overline{N_{K/F}(1 + \tilde{z}\varpi_K^t) - 1}}{\varpi_F^t}.$$

To determine the critical norm polynomial P , expand the norm as

$$N_{K/F}(1 + x) = 1 + \mathrm{Tr}_{K/F}(x) + s_2(x) + \cdots + s_{p-1}(x) + N_{K/F}(x),$$

where $s_j(x)$ is the j th elementary symmetric function of the conjugates of x . For $x = \tilde{z}\varpi_K^t$, the ramification estimates put all the terms $s_2(x), \dots, s_{p-1}(x)$ in $\mathfrak{p}_F^{t+1}$. Hence (30) gives

$$(31) \quad P(z) = z^p + cz, \quad c = \frac{\overline{\mathrm{Tr}_{K/F}(\varpi_K^t)}}{\varpi_F^t}.$$

The ramification map determines the kernel of P . Let σ be a generator of $\mathrm{Gal}(K/F) = G_t/G_{t+1}$, and put

$$\lambda = \frac{\overline{\sigma(\varpi_K) - \varpi_K}}{\varpi_K^{t+1}} \in k^\times.$$

The graded norm theorem [4, Theorem 3.1(ii)] identifies

$$\ker(U_K^t/U_K^{t+1} \longrightarrow U_F^t/U_F^{t+1})$$

with the image of the ramification map

$$G_t/G_{t+1} \longrightarrow U_K^t/U_K^{t+1}.$$

Under the identification $U_K^t/U_K^{t+1} \simeq k$, this map sends

$$\sigma^j \longmapsto j\lambda, \quad j \in \mathbf{F}_p.$$

Therefore

$$\ker(P) = \lambda \mathbf{F}_p.$$

In particular,

$$(32) \quad P(z) = \prod_{j \in \mathbf{F}_p} (z - j\lambda) = z^p - \lambda^{p-1}z.$$

Comparing (31) and (32) gives the normalized trace identity

$$(33) \quad \frac{\overline{\mathrm{Tr}_{K/F}(\varpi_K^t)}}{\varpi_F^t} = -\lambda^{p-1}.$$

In Lean, the theorem `criticalNormLinearCoefficient_eq_neg_ramificationLambda_pow` proves (33).

The image of the critical norm map also has a simple description. Writing $z = \lambda y$ in (32) gives

$$P(\lambda y) = \lambda^p(y^p - y).$$

The image of $y \mapsto y^p - y$ is the trace-zero subspace of k . Counting dimensions gives

$$(34) \quad \mathrm{im}(P) = \lambda^p \ker(\mathrm{Tr}_{k/\mathbf{F}_p} : k \rightarrow \mathbf{F}_p).$$

The Lean theorem `finiteField_scaledFrobeniusSub_range_eq_scaledTraceKer` proves exactly (34).

This image formula explains why the calculation is needed later. Let $\mu \in S(K/F)$ be nontrivial. The theorem `ramifiedNormCharacter_conductor` gives $m_F(\mu) = t + 1$. Thus the restriction of μ to U_F^t/U_F^{t+1} is a nontrivial additive character of k . Choose the standard additive character

$$(35) \quad \psi_0(z) = \exp\left(\frac{2\pi i}{p} \mathrm{Tr}_{k/\mathbf{F}_p}(z)\right).$$

There is then a unique $a \in k^\times$ such that

$$\mu(1 + \tilde{z} \varpi_F^t) = \psi_0(az) \quad (z \in k).$$

Because μ is trivial on norms, this character is trivial on the subspace in (34). The trace pairing on k is nondegenerate, and the annihilator of $\lambda^p \ker(\mathrm{Tr}_{k/\mathbf{F}_p})$ is the line $\lambda^{-p} \mathbf{F}_p$. Hence

$$a \in \lambda^{-p} \mathbf{F}_p^\times.$$

Thus every nontrivial $\mu \in S(K/F)$ has a unique $j \in \mathbf{F}_p^\times$ such that

$$(36) \quad \mu(1 + \tilde{z} \varpi_F^t) = \psi_0(j \lambda^{-p} z) \quad (z \in k).$$

This describes the restrictions of the nontrivial norm characters to U_F^t/U_F^{t+1} . We use it in the conductor-one calculation of Section 6.3 and the boundary calculation $m = T$ in Section 6.5.

5.3. Finite-field identities. The identities in this subsection are grouped according to the cases listed in Section 2.5. The conductor-zero parts of all cases use only the direct formulas on the local fields and therefore require no finite-field identity from this subsection. Some of these identities are used in more than one of the remaining cases.

5.3.1. Finite Gauss sums: the conductor-one cases. The finite-field formulas below are used in the conductor-one parts of the unramified and tamely ramified cases (Sections 6.1 and 6.2). The same finite Gauss sum also occurs in the wildly ramified conductor-one case in Section 6.3, but the comparison between the local constants over K and F is made by Frobenius rather than by a Hasse–Davenport formula.

Let k be a finite field. Lean uses `FiniteMulChar` for a multiplicative character of $k^\times$, extended by zero at 0, and `FiniteAddChar` for an additive character $k \rightarrow \mathbf{C}^\times$. For a multiplicative character χ and a nontrivial additive character ψ , Langlands’s normalization used throughout the proof is

$$(37) \quad \tau_k(\chi, \psi) = - \sum_{x \in k^\times} \chi(x)^{-1} \psi(x).$$

The Lean definition `langlandsGaussSum` is exactly the sum in (37). The inverse character and the initial minus sign are part of this normalization.

Let k'/k be a finite extension of degree f . The Hasse–Davenport lifting identity `hasseDavenportLift` is

$$(38) \quad \tau_{k'}(\chi \circ N_{k'/k}, \psi \circ \text{Tr}_{k'/k}) = \tau_k(\chi, \psi)^f.$$

This is used when a local Gauss sum over F is compared with the corresponding sum over an unramified extension.

The Hasse–Davenport product identity `hasseDavenportProduct` concerns a product of twists. Suppose that $\ell \mid |k^\times|$ and that η is a multiplicative character of order ℓ . Then

$$(39) \quad \chi(\ell^\ell) \tau_k(\chi^\ell, \psi) \prod_{j=1}^{\ell-1} \tau_k(\eta^j, \psi) = \tau_k(\chi, \psi) \prod_{j=1}^{\ell-1} \tau_k(\chi \eta^j, \psi),$$

where ℓ is viewed as an element of k . This identity is used after a conductor-one local constant has been reduced to a Gauss sum over the residue field.

5.3.2. Hasse functions: the odd higher-conductor cases. This subsection concerns the cases when the quasi-character in a local constant has odd conductor greater than one. In that case Lamprecht’s formula (24) contains the phase of a finite sum over the residue field. The translation and nonvanishing results below are used whenever a change of stationary representative changes the finite function: in Sections 6.2, 6.4, 6.5, and 6.6. The lifting identity (44) is used in the unramified case (Section 6.1).

Let k be a finite field and let $\eta : k \rightarrow \mathbf{C}^\times$ be an additive character. A Hasse function for η is a function $\phi : k \rightarrow \mathbf{C}^\times$ satisfying

$$(40) \quad \phi(x + y) = \phi(x)\phi(y)\eta(xy) \quad (x, y \in k).$$

The additive character η is therefore the polar character of ϕ . Lean represents this structure by `HasseFunction`.

Write

$$S(\phi) = \sum_{x \in k} \phi(x).$$

For $a \in k$, define another Hasse function for the same polar character by

$$\phi_a(x) := \phi(x)\eta(ax).$$

Summing over $x \in k$ and reindexing by $x \mapsto x + a$ therefore gives

$$(41) \quad S(\phi_a) = \phi(a)^{-1} S(\phi).$$

If η is nontrivial, then $S(\phi) \neq 0$. Taking phases in (41) gives

$$(42) \quad \text{ph}(S(\phi_a)) = \phi(a)^{-1} \text{ph}(S(\phi)).$$

The nonvanishing and the phase identity are proved by `HasseFunction.sum_ne_zero` and `HasseFunction.sumPhase_translate`. Equation (42) is used when changing a stationary representative translates the odd-conductor sum.

Assume that the polar character η is nontrivial. There is also a lifting identity for Hasse functions. Let k'/k be a finite extension of degree

$$f = [k' : k].$$

For $x \in k'$, put

$$\omega_{k'/k}(x) = \sum_{\{\sigma, \tau\}} \sigma(x)\tau(x) \in k,$$

where the sum is over unordered pairs of distinct k -embeddings of k' into an algebraic closure. Define

$$(43) \quad \phi_{k'/k}(x) := \phi(\text{Tr}_{k'/k}(x)) \eta(-\omega_{k'/k}(x)).$$

The Lean definition of this lift is `hasseFunctionLiftValue`. Then $\phi_{k'/k}$ is a Hasse function on k' for the additive character $\eta \circ \text{Tr}_{k'/k}$, and its sum satisfies

$$(44) \quad - \sum_{x \in k'} \phi_{k'/k}(x) = \left(- \sum_{x \in k} \phi(x) \right)^f.$$

The lifting formula (44) is proved by `hasseFunctionLift`. In the unramified case this identity compares the odd-conductor residue-field sum over k_K with the corresponding sum over k_F .

5.3.3. Quadratic phases: the tame quadratic and wild odd-prime cases. This subsection is used in the tamely ramified quadratic case and in both stable and non-stable wild odd-prime ranges: Sections 6.2, 6.4, and 6.5. It evaluates the residue-field sum occurring in Lamprecht's formula when the residue field has odd characteristic.

Let k have odd characteristic, and fix a nontrivial additive character

$$\psi : k \longrightarrow \mathbf{C}^\times.$$

In the applications, the polar character of the Hasse function has the form

$$\eta_a(z) = \psi(az) \quad \text{for some } a \in k^\times.$$

Consider the Hasse function $\Phi : k \rightarrow \mathbf{C}^\times$ for η_a :

$$\frac{\Phi(x+y)}{\Phi(x)\Phi(y)} = \psi(axy) \quad (x, y \in k).$$

There is a unique $b \in k$ such that

$$(45) \quad \Phi(x) = \psi\left(\frac{a}{2}x^2 + bx\right) \quad (x \in k).$$

The normal form (45) is proved by `CriticalPolarFunction.eq_quadratic`, and uniqueness of b by `CriticalPolarFunction.affineCoefficient_unique`.

Thus the finite sum in Lamprecht's formula is a quadratic Gauss sum. Define

$$(46) \quad Q_\psi(a, b) = \text{ph} \left(\sum_{x \in k} \psi\left(\frac{a}{2}x^2 + bx\right) \right).$$

This is `quadraticPhase`. Here the sum is nonzero by the Hasse-sum nonvanishing result in Section 5.3.2. Let $\nu : k^\times \rightarrow \{\pm 1\}$ be the quadratic character, defined by

$$(47) \quad \nu(x) = \begin{cases} 1, & x \text{ is a square in } k^\times, \\ -1, & x \text{ is a nonsquare in } k^\times. \end{cases}$$

Completing the square and applying the standard quadratic Gauss-sum identities give

$$(48) \quad Q_\psi(a, b) = \nu(a) \psi\left(-\frac{b^2}{2a}\right) Q_\psi(1, 0), \quad Q_\psi(1, 0)^2 = \nu(-1).$$

The two identities in (48) are proved by `quadraticPhase_eq_basic` and `quadraticPhase_basic_sq`. They evaluate the finite phases remaining in the tame quadratic and wildly ramified odd-prime calculations.

5.3.4. Ramified Hasse phases: the tame odd and stable wild odd-prime cases. This comparison is used in the higher-conductor tame odd-degree case and in the stable wild odd-prime case (Sections 6.2 and 6.4). Recall that Lamprecht's formula has no residue-field phase when the conductor is even.

Let K/F be totally ramified cyclic of odd prime degree ℓ , with lower break t . Write k for the common residue field and suppose

$$m_F(\chi_F) = 2d + 1 > 1.$$

Assume either $t = 0$, or that K/F is wild and $d \geq t + 1$. By (27), the norm pullback $\chi_K = \chi_F \circ N_{K/F}$ also has odd conductor, so Lamprecht's odd-conductor formula applies over both F and K .

Choose an admissible element $\gamma \in F^\times$ for both F and K . The stationary classes of χ_F and χ_K admit the same representative $\beta \in \mathcal{O}_F^\times$. Let H_F and H_K be the phases of the resulting residue-field sums. Then

$$(49) \quad H_K = H_F^\ell.$$

The norm calculation relates the two critical functions defining the phases in (49). In odd residue characteristic, one applies the quadratic-phase formulas (45) and (48); in characteristic two, one uses the Hasse-function identity (40). The remaining sign calculation uses $\ell \mid |k^\times|$ in the tame case. The full argument is [4, Proposition 5.6].

For compatible stationary representatives, the phase identity (49) is proved by `ramifiedHasseComparison`. The norm and finite-field calculations needed for this application are supplied by `ramifiedHassePreparation`.

5.3.5. The affine-pencil identity. The following finite-field identity is used at the boundary $m = T$ in Section 6.5.

Let k be a finite field of characteristic $p > 2$ and let $\psi : k \rightarrow \mathbf{C}^\times$ be a nontrivial additive character satisfying $\psi(z^p) = \psi(z)$ for all $z \in k$. The character ψ_0 defined in (35) has this property. Suppose $a_0, b_0, \rho, \vartheta, \sigma \in k$ satisfy

$$(50) \quad \begin{aligned} \rho &\neq 0, & 1 + j\rho &\neq 0 \quad (j \in \mathbf{F}_p), \\ a_0^p &= 1 - (\rho^{p-1})^{-1}, & b_0^p &= \sigma - \rho^{-1}\vartheta. \end{aligned}$$

Then

$$(51) \quad Q_\psi(a_0, b_0) \prod_{j \in \mathbf{F}_p^\times} Q_\psi(j\rho, j\vartheta) = \prod_{j \in \mathbf{F}_p} Q_\psi(1 + j\rho, \sigma + j\vartheta).$$

The hypotheses ensure that all quadratic coefficients are nonzero. The pairs $(1 + j\rho, \sigma + j\vartheta)$ form an affine line in k^2 , which explains the name *affine-pencil identity*. Its Lean theorem is `affinePencil`.

In the boundary case $m = T$ of Section 6.5, the coefficient pairs that actually occur have an extra common factor $c \in k^\times$: the first pair is (ca_0^p, cb_0^p) , and every other pair is multiplied by c . Proposition 5.8 of [4] shows that this common factor does not change the quotient of quadratic phases. Hence the boundary calculation reduces to the affine-pencil identity (51).

This identity is needed only when T is odd; when T is even, all the corresponding phases are 1.

5.3.6. Characteristic-two quadratic refinements. This subsection supplies the finite-field identities used in the wild quadratic case (Section 6.6).

Let k be a finite field of characteristic two, and put

$$\psi_0(z) = (-1)^{\text{Tr}_{k/\mathbf{F}_2}(z)}.$$

The quadratic expressions of Section 5.3.3 are unavailable in characteristic two. Instead, multiplying a Hasse function with polar character ψ_0 by a suitable additive character gives a quadratic refinement $\mathfrak{q} : k \rightarrow \mathbf{C}^\times$ satisfying

$$(52) \quad \mathfrak{q}(x + y) = \mathfrak{q}(x)\mathfrak{q}(y)\psi_0(xy), \quad \mathfrak{q}(x^2) = \mathfrak{q}(x).$$

If h is another Hasse function with the same polar character, then $h/\mathfrak{q}$ is an additive character. The perfect trace pairing therefore gives a unique $\lambda \in k$ such that $h = h_\lambda$, where

$$(53) \quad h_\lambda(x) = \mathfrak{q}(x)\psi_0(\lambda x), \quad \lambda \in k.$$

Write

$$H(\lambda) = \text{ph} \left(\sum_{x \in k} h_\lambda(x) \right), \quad H_0 = H(0).$$

Translation of the sum gives

$$(54) \quad H(\lambda) = H_0 \overline{\mathfrak{q}(\lambda)}, \quad H_0^2 = \mathfrak{q}(1), \quad \mathfrak{q}(x)^2 = \psi_0(x).$$

In particular,

$$(55) \quad H(\lambda)^2 = h_\lambda(1), \quad H(\lambda+1)H(\lambda) = 1, \quad H(\lambda^2) = H(\lambda).$$

Lean represents the normalized refinement by `CharTwoRefinement`. Its existence is proved by `CharTwoRefinement.exists_charTwoRefinement_translate`, and the unique parametrization by λ is proved by `CharTwoRefinement.existsUnique_affine`.

The boundary calculation uses the following identity for four such functions. Let $\rho \in k^\times$ with $1 + \rho \neq 0$, let $r^2 = \rho$, and suppose $\lambda_0, \lambda, \lambda_1, \lambda' \in k$ satisfy

$$(56) \quad \lambda_1 = \frac{\lambda_0 + \rho\lambda + r}{1 + \rho}, \quad (\lambda')^2 = \frac{\lambda + \rho\lambda_0 + \rho + r}{1 + \rho}.$$

Put

$$(57) \quad R = \frac{H(\lambda')H(\lambda_0)}{H(\lambda)H(\lambda_1)}, \quad B = \frac{\rho^2}{1 + \rho^2}, \quad C = \frac{\rho}{1 + \rho^2}.$$

Then

$$(58) \quad R h_\lambda(B) h_{\lambda_0}(C) \psi_0(C) = \psi_0(c + c^2) = 1, \quad c = (1 + \rho)^{-1}.$$

The first equality follows from the refinement identities (52) and (54); the last uses $\text{Tr}_{k/\mathbf{F}_2}(c + c^2) = 0$. Lean proves this cancellation by `WildQuadraticBoundaryData.value_eq_one`, with the last step supplied by `artinSchreier_phase`.

In Section 6.6, we identify the critical functions for the four local characters and verify these coefficient relations before applying the identity.

6. THE INDIVIDUAL CASES

For each case of the First Main Lemma (1), we explain the mathematical reduction and then give the Lean declaration proving the corresponding `FirstMainIdentity`. The reductions use the local-field factors of Section 4 and the ramification and finite-field results of Section 5.

Recall the setup and list of cases of Section 2.5. By the twist invariance (8), we may first replace χ_F by a character of minimal conductor in its $S(K/F)$ -orbit. Write

$$m = m_F(\chi_F) = 2d + \varepsilon, \quad \varepsilon \in \{0, 1\}.$$

In the ramified cases, let t be the lower ramification break and put

$$T = t + 1.$$

Each case below proves

$$\mathcal{E}_{K/F}(\chi_F) = 1,$$

which is equivalent to the First Main Lemma (1).

Let E be a nonarchimedean local field with residue field k_E , $\theta : E^\times \rightarrow \mathbf{C}^\times$ a quasi-character, and $\psi_E : E \rightarrow \mathbf{C}^\times$ a nontrivial additive character. Choose a uniformizer ϖ_E of E .

Formulas for conductors zero and one. If θ has conductor at most one, its residue character is

$$\bar{\theta}(\bar{u}) = \theta(\tilde{u}), \quad \bar{u} \in k_E^\times.$$

Put $n_E = n_E(\psi_E)$. For $v_E(\gamma) = n_E + 1$, define

$$\bar{\psi}_{E,\gamma}(\bar{x}) = \psi_E(\tilde{x}/\gamma).$$

Here tildes denote arbitrary lifts. The character $\bar{\psi}_{E,\gamma}$ is a nontrivial additive character of k_E . When $\gamma = \varpi_E^{n_E+1}$, write $\bar{\psi}_{E,\gamma} = \bar{\psi}_E$. The definition of Δ gives

$$(59) \quad \Delta(\theta, \psi_E) = \begin{cases} \theta(\varpi_E)^{n_E}, & m_E(\theta) = 0, \\ \theta(\gamma) \text{ph}(-\tau_{k_E}(\bar{\theta}, \bar{\psi}_{E,\gamma})), & m_E(\theta) = 1. \end{cases}$$

The minus sign comes from the convention (37). These formulas are formalized by `deltaFinite_conductor_zero` and `deltaFinite_conductor_one`.

Stable twists. Let $\omega : E^\times \rightarrow \mathbf{C}^\times$ be a quasi-character. Write $m_E(\theta) = 2d + \varepsilon > 1$, where $d \geq 0$ is an integer and $\varepsilon \in \{0, 1\}$, and suppose $m_E(\omega) \leq d$. Choose an admissible element γ and a stationary representative β for θ . Then $\omega\theta$ has the same conductor as θ , the same γ is admissible, and the same β is a stationary representative. If the conductor is odd, choose the same critical element δ for both characters. Their critical functions then agree. The stable-twist formula is

$$(60) \quad \Delta(\omega\theta, \psi_E) = \omega(\gamma/\beta)\Delta(\theta, \psi_E).$$

Its Lean counterpart is `stableTwist`.

Products over norm characters. For a finite-order character $\omega : E^\times \rightarrow \mathbf{C}^\times$, [4, Lemma 2.11] gives

$$\Delta(\omega, \psi_E)\Delta(\omega^{-1}, \psi_E) = \omega(-1).$$

Specialize to $E = F$. If ℓ is odd, then every $\mu \in S(K/F)$ has odd order, so $\mu(-1) = 1$. Pairing each nontrivial μ with μ^{-1} , and using $\Delta(1, \psi_F) = 1$, gives

$$(61) \quad \prod_{\mu \in S(K/F)} \Delta(\mu, \psi_F) = 1 \quad (\ell \text{ odd}).$$

The inverse identity and the product formula are formalized by `delta_inverse_pair` and `delta_odd_prime_character_product`.

Ramified conductor zero. Suppose that K/F is ramified of prime degree ℓ and $m = 0$. Choose a uniformizer ϖ_K of K , put $\varpi_F = N_{K/F}(\varpi_K)$, and write $n = n_F(\psi_F)$. For each $\mu \in S(K/F)$, the finite-sum formula (5) and the fact that χ_F is unramified give

$$\Delta(\mu\chi_F, \psi_F) = \chi_F(\varpi_F)^{m_F(\mu)+n}\Delta(\mu, \psi_F).$$

Each nontrivial norm character has conductor T , so

$$\prod_{\mu \in S(K/F)} \Delta(\mu\chi_F, \psi_F) = \chi_F(\varpi_F)^{n+(\ell-1)(T+n)} \prod_{\mu \in S(K/F)} \Delta(\mu, \psi_F).$$

By (26),

$$n + (\ell - 1)(T + n) = \ell n + (\ell - 1)T = n_K(\psi_K).$$

By (28), $m_K(\chi_K) = 0$. So (59) gives

$$\Delta(\chi_K, \psi_K) = \chi_K(\varpi_K)^{n_K(\psi_K)} = \chi_F(N_{K/F}(\varpi_K))^{n_K(\psi_K)} = \chi_F(\varpi_F)^{n_K(\psi_K)}.$$

Hence

$$\Delta(\chi_K, \psi_K) \prod_{\mu \in S(K/F)} \Delta(\mu, \psi_F) = \prod_{\mu \in S(K/F)} \Delta(\mu\chi_F, \psi_F),$$

which is the First Main Lemma when $m = 0$.

6.1. Unramified extensions. Suppose that K/F is unramified of prime degree ℓ . Choose a uniformizer ϖ of F , which is also a uniformizer of K , and put $n = n_F(\psi_F)$. The norm is surjective on units, so every $\mu \in S(K/F)$ is unramified. The finite-sum formula (5) gives

$$\Delta(\mu\chi_F, \psi_F) = \mu(\varpi)^{m+n}\Delta(\chi_F, \psi_F), \quad \Delta(\mu, \psi_F) = \mu(\varpi)^n.$$

Taking products over $S(K/F)$ therefore reduces the First Main Lemma to

$$(62) \quad \Delta(\chi_K, \psi_K) = (-1)^{m(\ell-1)}\Delta(\chi_F, \psi_F)^\ell.$$

For $m = 0$, both χ_F and χ_K are unramified. By (59), we have

$$\Delta(\chi_F, \psi_F) = \chi_F(\varpi)^n, \quad \Delta(\chi_K, \psi_K) = \chi_K(\varpi)^n.$$

Since $\chi_K = \chi_F \circ N_{K/F}$ and $N_{K/F}(\varpi) = \varpi^\ell$, (62) follows.

For $m = 1$, let k_K/k_F be the residue field extension, and let $\bar{\chi}$ and $\bar{\psi}$ be the residue characters over F . Since K/F is unramified, (29) gives $n_K(\psi_K) = n$, and the residue characters over K are

$$\bar{\chi}_K = \bar{\chi} \circ N_{k_K/k_F}, \quad \bar{\psi}_K = \bar{\psi} \circ \text{Tr}_{k_K/k_F}.$$

Both χ_F and χ_K have conductor one by (29). Using $\chi_K(\varpi) = \chi_F(\varpi)^\ell$, the conductor-one formula (59) gives

$$\begin{aligned}\Delta(\chi_F, \psi_F) &= \chi_F(\varpi)^{n+1} \text{ph}\left(-\tau_{k_F}(\bar{\chi}, \bar{\psi})\right), \\ \Delta(\chi_K, \psi_K) &= \chi_F(\varpi)^{\ell(n+1)} \text{ph}\left(-\tau_{k_K}(\bar{\chi}_K, \bar{\psi}_K)\right).\end{aligned}$$

The Hasse–Davenport lifting formula (38) states that

$$\tau_{k_K}(\bar{\chi}_K, \bar{\psi}_K) = \tau_{k_F}(\bar{\chi}, \bar{\psi})^\ell.$$

Substituting and taking phases yields

$$\Delta(\chi_K, \psi_K) = (-1)^{\ell-1} \Delta(\chi_F, \psi_F)^\ell,$$

which is (62) for $m = 1$.

For $m > 1$, put $\gamma = \varpi^{m+n}$. By (29), $m_K(\chi_K) = m$ and $n_K(\psi_K) = n$, so γ is admissible over both fields. Choose a stationary representative $\beta \in \mathcal{O}_F^\times$ for χ_F relative to γ . For $y \in \mathfrak{p}_K^{\lceil m/2 \rceil}$, we have

$$P_{K/F}(y) \equiv \text{Tr}_{K/F}(y) \pmod{\mathfrak{p}_F^m},$$

and hence

$$\chi_K(1+y) = \psi_F\left(\frac{\beta \text{Tr}_{K/F}(y)}{\gamma}\right) = \psi_K\left(\frac{\beta y}{\gamma}\right).$$

Thus (14) shows that the same β represents the stationary class of χ_K .

Since $\beta, \gamma \in F$, the norm and trace identities give the relation between local-field factors in (13):

$$(63) \quad \mathcal{A}_K(\chi_K; \beta, \gamma) = \mathcal{A}_F(\chi_F; \beta, \gamma)^\ell.$$

If m is even, Lamprecht’s formula has no residue-field phase, so this proves (62).

If $m = 2d + 1$, Lamprecht’s formula gives

$$\Delta(\chi_E, \psi_E) = \mathcal{A}_E(\chi_E; \beta, \gamma) \text{ph}\left(\sum_{x \in k_E} \phi_{\chi_E, \beta}(x)\right).$$

To compare the two sums, we may take $\delta = \varpi^d$ in both critical functions. Expanding $N_{K/F}(1 + \delta \tilde{x})$ in (11) to second order identifies $\phi_{\chi_K, \beta}$ with the Hasse lift of $\phi_{\chi_F, \beta}$ defined in (43). The lifting formula (44) therefore gives

$$\sum_{x \in k_K} \phi_{\chi_K, \beta}(x) = (-1)^{\ell-1} \left(\sum_{x \in k_F} \phi_{\chi_F, \beta}(x) \right)^\ell.$$

Taking phases and combining it with (63) proves (62).

The Lean theorem for this case is `firstMain_unramified_dispatchReady`.

6.2. Tamely ramified extensions. In the tamely ramified case, F and K have the same residue field k , and the lower ramification break is $t = 0$. Every nontrivial member of $S(K/F)$ has conductor one.

The conductor-zero case was proved at the beginning of this section. Its Lean counterparts are `firstMain_tameOdd_zero` for odd degree and `firstMain_tameQuadratic_conductor_zero` for degree two.

For the remaining cases, pick a uniformizer ϖ_K of K , put $\varpi_F = N_{K/F}(\varpi_K)$, and write $n = n_F(\psi_F)$.

Suppose that $m = 1$ and choose a generator τ of $S(K/F)$. Every nontrivial τ^j has conductor one. Since χ_F has minimal conductor in its $S(K/F)$ -orbit, every twist $\tau^j \chi_F$ also has conductor one. Moreover, (28) gives

$$m_K(\chi_K) = 1.$$

Let $\bar{\chi}$ and η be the residue characters of χ_F and τ . Then η has order ℓ .

Put $\gamma = \varpi_F^{n+1}$. It is admissible for every conductor-one character over F . Moreover, by (26),

$$v_K(\gamma) = \ell(n+1) = n_K(\psi_K) + 1,$$

so the same element is admissible for χ_K over K . Put $\bar{\psi} = \bar{\psi}_F$. The residue character of χ_K is $\bar{\chi}^\ell$, and the residue character associated to ψ_K using γ is

$$x \mapsto \bar{\psi}(\ell x).$$

The scaling identity

$$\tau_k(\alpha, \bar{\psi}(\ell \cdot)) = \alpha(\ell) \tau_k(\alpha, \bar{\psi})$$

therefore contributes $\bar{\chi}(\ell^\ell)$ to the local constant over K . Since γ is a power of the norm ϖ_F , we have $\tau(\gamma) = 1$. Also $\chi_K(\gamma) = \chi_F(\gamma)^\ell$. Thus the character factors in (59) agree on the two sides of (1); each side also has ℓ minus signs. The desired identity (1) follows by taking phases in

$$\bar{\chi}(\ell^\ell) \tau_k(\bar{\chi}^\ell, \bar{\psi}) \prod_{j=1}^{\ell-1} \tau_k(\eta^j, \bar{\psi}) = \tau_k(\bar{\chi}, \bar{\psi}) \prod_{j=1}^{\ell-1} \tau_k(\bar{\chi} \eta^j, \bar{\psi}).$$

This is the Hasse–Davenport multiplication formula (39).

Now assume that ℓ is odd and $m = 2d + \varepsilon > 1$. Every nontrivial $\mu \in S(K/F)$ has conductor one, so (60) applies. Since $\prod_{\mu \in S(K/F)} \mu = 1$, it gives

$$\prod_{\mu \in S(K/F)} \Delta(\mu \chi_F, \psi_F) = \Delta(\chi_F, \psi_F)^\ell.$$

By (61), the First Main Lemma then reduces to $\Delta(\chi_K, \psi_K) = \Delta(\chi_F, \psi_F)^\ell$.

Put $\gamma = \varpi_F^{m+n}$. By (26) and (27),

$$m_K(\chi_K) + n_K(\psi_K) = \ell(m+n) = v_K(\gamma),$$

so γ is admissible over both F and K . Since $t = 0$ and $d \geq 1 = t + 1$, [4, Proposition 6.8(b)] shows that the stationary classes of χ_F and χ_K admit the same representative $\beta \in \mathcal{O}_F^\times$. The norm and trace identities on F give

$$\mathcal{A}_K(\chi_K; \beta, \gamma) = \mathcal{A}_F(\chi_F; \beta, \gamma)^\ell.$$

For even conductor this proves the required identity (1). For odd conductor, (49) gives the same power relation for the residue-field phases, proving (1). The corresponding Lean theorem is `firstMain_tameOdd_of_isDeltaFinite`.

It remains to treat the tamely ramified quadratic case with $m = 2d + \varepsilon \geq 2$. Write $S(K/F) = \{1, \tau\}$ and $n = n_F(\psi_F)$, and let ν be the quadratic character (47), which is the residue character of τ . Choose uniformizers satisfying $\varpi_F = N_{K/F}(\varpi_K) = -\varpi_K^2$.

Take $\gamma = \varpi_F^{m+n}$. The conductor formulas (26) and (27) give again $v_K(\gamma) = m_K(\chi_K) + n_K(\psi_K)$, so γ is admissible over both fields.

Since τ has conductor one, it is trivial on U_F^d , so χ_F and $\tau \chi_F$ have the same stationary class. By [4, Proposition 6.8(d)], the stationary class of χ_K has the same representative. Thus we may choose $\beta \in \mathcal{O}_F^\times$ as a common stationary representative for all.

Note that $\tau(\gamma) = 1$ because it is trivial on norms. Hence the stable-twist formula (60) gives

$$\Delta(\tau \chi_F, \psi_F) = \nu(\bar{\beta}) \Delta(\chi_F, \psi_F).$$

Moreover, $N_{K/F}(a) = a^2$ and $\text{Tr}_{K/F}(a) = 2a$ for $a \in F$, so the local-field factors in Lamprecht's formula satisfy $\mathcal{A}_K = \mathcal{A}_F^2$. Thus it remains to prove

$$(64) \quad H_K \Delta(\tau, \psi_F) = \nu(\bar{\beta}) H_F^2,$$

where H_E is the phase of the residue-field sum for χ_E , with $H_E = 1$ when $m_E(\chi_E)$ is even.

Throughout the following calculation, fix the additive character

$$\bar{\psi} = \bar{\psi}_F$$

of the common residue field k , and put $Q_0 = Q_{\bar{\psi}}(1, 0)$. All quadratic functions below are expressed using this same $\bar{\psi}$, including those arising over K . For each function $x \mapsto \bar{\psi}(\frac{a}{2}x^2 + bx)$, its sum has phase $Q_{\bar{\psi}}(a, b)$ by (46), and (48) gives

$$Q_{\bar{\psi}}(a, b) = \nu(a)\bar{\psi}\left(-\frac{b^2}{2a}\right)Q_0, \quad Q_0^2 = \nu(-1).$$

First consider τ . Using

$$\sum_{y \in k^\times} \nu(y)\bar{\psi}(y) = \sum_{x \in k} \bar{\psi}(x^2),$$

the conductor-one formula (59) gives

$$(65) \quad \Delta(\tau, \psi_F) = Q_{\bar{\psi}}(2, 0) = \nu(2)Q_0.$$

If $m = 2d$, then $H_F = 1$. Taking $\delta_K = \varpi_K^{2d-1}$ in the critical function over K gives

$$\phi_{\chi_K, \beta}(x) = \bar{\psi}(-\bar{\beta}x^2).$$

Thus

$$H_K = Q_{\bar{\psi}}(-2\bar{\beta}, 0) = \nu(-2\bar{\beta})Q_0.$$

Combining this with (65) proves (64) in this case.

If $m = 2d + 1$, take $\delta_F = \varpi_F^d$. By (12), the polar character of $\phi_{\chi_F, \beta}$ is $z \mapsto \bar{\psi}(\bar{\beta}z)$. Applying (45) with $\psi = \bar{\psi}$ and $a = \bar{\beta}$ gives a unique $c \in k$ such that

$$\phi_{\chi_F, \beta}(x) = \bar{\psi}\left(\frac{\bar{\beta}}{2}x^2 + cx\right).$$

For the critical function over K , choose $\delta_K = \varpi_K^{2d} = (-1)^d \delta_F$. The norm and trace identities then give

$$\phi_{\chi_K, \beta}(x) = \phi_{\chi_F, \beta}((-1)^d x)^2 = \bar{\psi}(\bar{\beta}x^2 + 2(-1)^d cx).$$

Applying (48) separately to these two functions yields

$$H_F = Q_{\bar{\psi}}(\bar{\beta}, c) = \nu(\bar{\beta})\bar{\psi}\left(-\frac{c^2}{2\bar{\beta}}\right)Q_0,$$

$$H_K = Q_{\bar{\psi}}(2\bar{\beta}, 2(-1)^d c) = \nu(2\bar{\beta})\bar{\psi}\left(-\frac{c^2}{\bar{\beta}}\right)Q_0.$$

Multiplying the second formula by $\nu(2)Q_0$ from (65) and using $Q_0^2 = \nu(-1)$ proves (64).

Lean proves the complete tame quadratic case in `firstMain_tameQuadratic_of_isDeltaFinite`.

6.3. Wild conductors zero and one. Suppose that K/F is wildly ramified of prime degree p (including the case $p = 2$), with lower break t and $T = t + 1$. Take uniformizers with $\varpi_F = N_{K/F}(\varpi_K)$.

The conductor-zero case is the common ramified calculation at the beginning of this section. Its Lean counterpart is `firstMain_wildEndpoint_zero`.

Now let $m = 1$. By (28), $m_K(\chi_K) = 1$.

Choose an admissible element γ_F whose residue additive character is ψ_0 from (35). Let f be the minimal polynomial of ϖ_K and put

$$\gamma_K = \gamma_F \frac{f'(\varpi_K)}{\varpi_K^{p-1}}.$$

Since $v_K(f'(\varpi_K)) = (p-1)T$, (26) gives $v_K(\gamma_K) = n_K(\psi_K) + 1$, so γ_K is admissible. The identity

$$\mathrm{Tr}_{K/F}\left(\frac{\varpi_K^{p-1}}{f'(\varpi_K)}\right) = 1,$$

together with the trace-ideal formula (25), shows that the residue additive character defined using γ_K is also ψ_0 .

The residue characters of χ_F and χ_K are $\bar{\chi}$ and $x \mapsto \bar{\chi}(x^p)$. Since $\psi_0(x^p) = \psi_0(x)$, Frobenius identifies their Gauss sums. Thus (59) gives

$$(66) \quad \frac{\Delta(\chi_K, \psi_K)}{\Delta(\chi_F, \psi_F)} = \chi_F \left(\frac{N_{K/F}(\gamma_K)}{\gamma_F} \right).$$

Let τ generate $S(K/F)$. Choose an admissible element γ_τ and a stationary representative β_τ , and put $c_\tau = \gamma_\tau / \beta_\tau$. The stable-twist formula (60) applies with $\theta = \tau^j$ and $\omega = \chi_F$, giving

$$\frac{\Delta(\tau^j \chi_F, \psi_F)}{\Delta(\tau^j, \psi_F)} = \chi_F(c_\tau / j), \quad 1 \leq j < p.$$

Since χ_F is trivial on U_F^1 and $\prod_{j=1}^{p-1} j = (-1)^p$ in the residue field k , multiplication gives

$$(67) \quad \prod_{j=1}^{p-1} \frac{\Delta(\tau^j \chi_F, \psi_F)}{\Delta(\tau^j, \psi_F)} = \chi_F((-1)^p c_\tau^{p-1}).$$

Restricting $\tau(1+x) = \psi_F(x/c_\tau)$ to $x = \varpi_F^t \tilde{z}$ and using $\psi_0(\bar{y}) = \psi_F(y/\gamma_F)$ give

$$\tau(1 + \varpi_F^t \tilde{z}) = \psi_0 \left(\frac{\overline{\varpi_F^t \gamma_F}}{c_\tau} z \right).$$

By (36), $\frac{\overline{\varpi_F^t \gamma_F}}{c_\tau} \in \lambda^{-p} \mathbf{F}_p^\times$, hence

$$\left(\frac{\overline{c_\tau}}{\overline{\varpi_F^t \gamma_F}} \right)^{p-1} = \lambda^{p(p-1)}.$$

On the other hand, for

$$D' = \frac{f'(\varpi_K)}{\varpi_K^{(p-1)T}} \in \mathcal{O}_K^\times,$$

[4, Lemma 9.1] gives

$$\overline{N_{K/F}(D')} = (-1)^p \lambda^{p(p-1)}.$$

We derive easily

$$(68) \quad \frac{N_{K/F}(\gamma_K)}{\gamma_F} \equiv (-1)^p c_\tau^{p-1} \pmod{U_F^1}.$$

Because χ_F is trivial on U_F^1 , (68) identifies its values in (66) and (67). Together with $\Delta(1, \psi_F) = 1$, this proves the First Main Lemma. Its Lean counterpart is `firstMain_wildEndpoint_one`.

6.4. The stable wild odd-prime range. Assume that K/F is wildly ramified of prime degree $p > 2$ and that $m = 2d + \varepsilon \geq 2T$, so $d \geq T = t + 1$. Every nontrivial norm character has conductor $T \leq d$. The stable-twist formula (60), the identity $\prod_{\mu \in S(K/F)} \mu = 1$, and (61) reduce the First Main Lemma, just as in the tame odd-degree case, to

$$(69) \quad \Delta(\chi_K, \psi_K) = \Delta(\chi_F, \psi_F)^p.$$

Choose an admissible element $\gamma \in F^\times$ for (χ_F, ψ_F) . The conductor formulas (26) and (27) give

$$v_K(\gamma) = p(m + n_F(\psi_F)) = m_K(\chi_K) + n_K(\psi_K).$$

So γ is also admissible for (χ_K, ψ_K) . Since $d \geq T = t + 1$, [4, Proposition 6.8(b)] allows us to choose a common stationary representative $\beta \in \mathcal{O}_F^\times$ for χ_F and χ_K with this γ .

Because $\beta, \gamma \in F$, the norm and trace identities give

$$\begin{aligned} \mathcal{A}_K(\chi_K; \beta, \gamma) &= \chi_F(\gamma^p) \psi_F(p\beta/\gamma) \chi_F(\beta^p)^{-1} \\ &= \mathcal{A}_F(\chi_F; \beta, \gamma)^p. \end{aligned}$$

Since p is odd, the conductor formula (27) also gives $m_K(\chi_K) \equiv m \pmod{2}$. If both conductors are even, the residue-field factors in Lamprecht's formula are both 1. If they are odd, (49) gives $\mathcal{H}_K = \mathcal{H}_F^p$. Together with the equality for $\mathcal{A}_K$, Lamprecht's formula (13) therefore proves (69).

The corresponding Lean theorem is `firstMain_wildStable_of_isDeltaFinite`.

6.5. The non-stable wild odd-prime range. Assume that K/F is wildly ramified of prime degree $p > 2$, with common residue field k , and that

$$1 < m = m_F(\chi_F) = 2d + \varepsilon < 2T, \quad T = t + 1.$$

Put

$$b = m - 1.$$

Choose a generator τ of $S(K/F)$, so that

$$S(K/F) = \{\tau^j : j \in \mathbf{F}_p\}.$$

For $j \in \mathbf{F}_p^\times$, the character τ^j has conductor T . Since $T > d$, the stable-twist argument of Section 6.4 does not apply: the stationary classes of τ^j , $\tau^j \chi_F$, and $\chi_K = \chi_F \circ N_{K/F}$ must be compared explicitly.

For each nontrivial norm character τ^j , we compare the stationary representatives used in the local-field and finite-field calculations. When $m_F(\chi_F) = T$, we also compare these representatives for $\tau^j \chi_F$. Denote the two representatives for a character θ by $\beta_\theta^{\text{loc}}$ and β_θ^{ff} , respectively.

When $m_F(\theta) = 2d_\theta + 1$, these choices satisfy

$$(70) \quad \beta_\theta^{\text{loc}} - \beta_\theta^{\text{ff}} \in \mathfrak{p}_F^{d_\theta+1}.$$

The critical functions use the same admissible element and the same critical element, so they agree. The congruence is proved for $b > t$ and $b \leq t$ by `wildOdd_highRepresentativeDifference_mem_succ` and `wildOdd_lowRepresentativeDifference_mem_succ`, respectively.

For $j \neq 0$, let R_j be the phase in Lamprecht's formula for τ^j . Write T_j for the phase for $\tau^j \chi_F$ and U for the phase for χ_K . The quotient of these phases in the error term (7) is

$$(71) \quad \mathcal{R}_{\text{odd}} := U \frac{\prod_{j \in \mathbf{F}_p^\times} R_j}{\prod_{j \in \mathbf{F}_p} T_j}.$$

For the chosen representatives, each odd-conductor factor is a quadratic phase as in Section 5.3.3. The congruence (70) ensures that the change of representative introduces no extra phase. This comparison is proved in `Cases/WildOdd/PhasePreparation.lean`.

The quotient (71) is evaluated according to the position of b relative to t . If $b < t$, then

$$T_j = R_j \quad (j \neq 0), \quad T_0 = U,$$

so all phases cancel pairwise. If $b > t$, the quadratic-phase formula of Section 5.3.3 gives

$$U = \prod_{j \in \mathbf{F}_p} T_j, \quad \prod_{j \in \mathbf{F}_p^\times} R_j = 1.$$

These nonboundary identities are formalized in `Cases/WildOdd/ScalarPhaseProducts.lean`.

At the boundary $b = t$, equivalently $m = T$, the characters in the Lamprecht comparison all have conductor T ; for χ_K , this follows from (28). If T is even, every residue-field phase is 1. Suppose now that T is odd, and use the additive character ψ_0 defined in (35).

Since $m = T$, we may use the same admissible element γ and critical element δ for τ and χ_F . With stationary representatives β_τ and β_{χ_F} , use $\beta_{\chi_F} + j\beta_\tau$ for $\tau^j \chi_F$, where in $j\beta_\tau$ we use the representative of $j \in \mathbf{F}_p$ in $\{0, \dots, p-1\}$. The definition (11) then gives

$$\phi_{\tau^j \chi_F, \beta_{\chi_F} + j\beta_\tau}(x) = \phi_{\chi_F, \beta_{\chi_F}}(x) \phi_{\tau, \beta_\tau}(x)^j.$$

Write the quadratic and linear coefficient pairs of $\phi_{\chi_F, \beta_{\chi_F}}$ and ϕ_{τ, β_τ} , relative to ψ_0 , as $(c, c\sigma)$ and $(c\rho, c\vartheta)$, respectively, where $c, \rho \in k^\times$. Consequently,

$$\begin{aligned} R_j &= Q_{\psi_0}(cj\rho, cj\vartheta) & (j \in \mathbf{F}_p^\times), \\ T_j &= Q_{\psi_0}(c(1+j\rho), c(\sigma+j\vartheta)) & (j \in \mathbf{F}_p). \end{aligned}$$

Minimality prevents the conductor of any twist from dropping, so $1+j\rho \neq 0$ for every $j \in \mathbf{F}_p$.

For the chosen representatives, the norm calculation in [4, Proposition 9.12] gives the coefficient pair

$$(c(1-\rho^{1-p}), c(\sigma-\rho^{-1}\vartheta))$$

for the critical function of χ_K . Thus, with a_0, b_0 as in (50),

$$U = Q_{\psi_0}(ca_0^p, cb_0^p).$$

These are exactly the scaled coefficient pairs considered at the end of Section 5.3.5. The common-factor version of the affine-pencil identity (51) therefore gives

$$U \prod_{j \in \mathbf{F}_p^\times} R_j = \prod_{j \in \mathbf{F}_p} T_j,$$

and hence $\mathcal{R}_{\text{odd}} = 1$. The Lean theorem is `wildOdd_boundaryPencil`.

Thus, in every case,

$$(72) \quad \mathcal{R}_{\text{odd}} = 1.$$

Lean combines all cases in `wildOdd_completeResidualPhaseQuotient_eq_one`.

It remains to compare the local-field factors in Lamprecht's formula.

Write β_θ and γ_θ for the chosen stationary representative and admissible element for $\theta \in \{\tau, \chi_F, \chi_K\}$. Both the $b > t$ and $b \leq t$ calculations give elements $A \in F^\times$ and $u \in K^\times$, with $n_u = N_{K/F}(u)$, such that

$$(73) \quad \frac{\beta_\tau}{\gamma_\tau} = A, \quad \frac{\beta_{\chi_F}}{\gamma_{\chi_F}} = An_u, \quad \frac{\beta_{\chi_K}}{\gamma_{\chi_K}} = A(n_u - u).$$

and

$$v_K(u) = v_F(n_u) = t - b.$$

By (73) and the definition of the stationary classes,

$$\begin{aligned} \tau(1+x) &= \psi_F(Ax) & (x \in \mathfrak{p}_F^{[T/2]}), \\ \chi_F(1+x) &= \psi_F(An_u x) & (x \in \mathfrak{p}_F^{[m/2]}). \end{aligned}$$

Substituting these relations and representatives into Lamprecht's formula gives

$$(74) \quad \mathcal{E}_{K/F}(\chi_F) = \mathcal{R}_{\text{odd}} \psi_F(-AX),$$

where $X \in F$ is the norm-trace correction computed below.

Since τ has conductor T , its stationary identity gives

$$(75) \quad \psi_F(Ax) = \tau(1+x) = 1 \quad (x \in \mathfrak{p}_F^T).$$

Thus it suffices to prove $X \in \mathfrak{p}_F^T$.

To compute X , let $s_r = s_r(u)$ be the r th elementary symmetric function of the conjugates of u , so that $s_1 = \text{Tr}_{K/F}(u)$ and $s_p = n_u$. The norm calculation gives the exact identity

$$(76) \quad X = \frac{pB}{1 - n_u^{p-1}}, \quad B = s_1 + \sum_{r=1}^{p-2} (-n_u)^r s_{p-r}.$$

The denominator is nonzero: when $b \neq t$, this follows from $v_F(n_u) = t - b$; at the boundary $b = t$, it follows from the minimality of χ_F in its $S(K/F)$ -orbit.

This formula treats equal and mixed characteristic separately. If $\text{char } F = p$, then $p = 0$ in F , so $X = 0$ by (76). If $\text{char } F = 0$, the ramification bounds for p and the symmetric functions $s_r(u)$,

together with the valuation of the denominator, give $X \in \mathfrak{p}_F^T$. In either case $\psi_F(-AX) = 1$. Combining this with $\mathcal{R}_{\text{odd}} = 1$ from (72) in (74) proves

$$\mathcal{E}_{K/F}(\chi_F) = 1.$$

The stationary representatives $\beta_\tau, \beta_\chi, \beta_{\chi_K}$ are chosen in `WildOdd/UpperResidualNormalization.lean` and `WildOdd/LowSourceNormalization.lean`. The required character linearizations are supplied by `Parameters/HighIntermediate.lean` for $b > t$, and by `WildOdd/CorrectionPreparation.lean` for $b \leq t$. The substitution giving (74) is proved in `Parameters/PhaseReduction.lean`. The exact correction formula is proved by `wildOdd_correctionFormula`.

The complete mathematical result is [4, Corollary 9.21]. Its Lean counterpart is `firstMain_wildOdd_nonstable`.

6.6. The wildly ramified quadratic extensions. Assume that K/F is wildly ramified quadratic, with $S(K/F) = \{1, \tau\}$, and put

$$m = m_F(\chi_F), \quad b = m - 1, \quad T = t + 1.$$

Write $k = k_F = k_K$ for the common residue field. The cases $m \leq 1$ were proved in Section 6.3, so assume $m > 1$.

Choose the stationary representatives β_θ and admissible elements γ_θ in [4, proof of Proposition 9.29], for $\theta \in \{\tau, \chi_F, \tau\chi_F, \chi_K\}$. For suitable $A \in F^\times$ and $u \in K^\times$, these choices satisfy

$$\begin{aligned} \frac{\beta_\tau}{\gamma_\tau} &= A, & \frac{\beta_{\chi_F}}{\gamma_{\chi_F}} &= An_u, \\ \frac{\beta_{\tau\chi_F}}{\gamma_{\tau\chi_F}} &= A(1 + n_u), & \frac{\beta_{\chi_K}}{\gamma_{\chi_K}} &= A(n_u - u), \end{aligned}$$

where $n_u = N_{K/F}(u)$ and $v_K(u) = v_F(n_u) = t - b$. The element $1 + n_u$ is nonzero by valuation when $b \neq t$, and by the minimality of χ_F when $b = t$.

For the odd-conductor characters, use the residue-field coordinates of [4, Proposition 9.27], with common polar character ψ_0 . Write the critical functions as $h_{\lambda_0}, h_\lambda, h_{\lambda_1}, h_{\lambda'}$ for $\tau, \chi_F, \tau\chi_F, \chi_K$, respectively, in the notation of (53). Let H_θ be the phase of the corresponding sum over k , with $H_\theta = 1$ when the conductor is even, and put

$$R_H = \frac{H_{\chi_K} H_\tau}{H_{\chi_F} H_{\tau\chi_F}}.$$

These choices are recorded in `WildQuadraticSelectedStationaryPairs`.

Put $s = \text{Tr}_{K/F}(u)$. Substituting the chosen representatives into the four Lamprecht formulas and using $N_{K/F}(1 \pm u) = 1 \pm s + n_u$ gives

$$(77) \quad \mathcal{E}_{K/F}(\chi_F) = R_H \psi_F(-As) \chi_F \left(1 - \frac{s}{1 + n_u}\right)^{-1} \tau \left(1 + \frac{s}{1 + n_u}\right)^{-1}.$$

The stationary identities and the critical-function formula (11) give the additive factors

$$\psi_F\left(\frac{An_us}{1 + n_u}\right), \quad \psi_F\left(-\frac{As}{1 + n_u}\right)$$

from the χ_F - and τ -evaluations, respectively. Their product with $\psi_F(-As)$ is $\psi_F(-AX)$, where

$$(78) \quad X = s - \frac{n_us}{1 + n_u} + \frac{s}{1 + n_u} = \frac{2s}{1 + n_u}.$$

The remaining factors are critical-function values. Their product with R_H is evaluated in [4, Proposition 9.29], as follows.

Outside the odd boundary. Suppose that $b \neq t$ or T is even, and put $w = s/(1 + n_u)$.

First consider the three cases

$$b < t, \quad b = t \text{ with } T \text{ even}, \quad b > t \text{ with } m \text{ even}.$$

For $b < t$, the coefficient relations in [4, Proposition 9.28], together with (55), give $H_{\chi_K} = H_{\chi_F}$ and $H_{\tau\chi_F} = H_\tau$. For $b = t$ with T even, all four phases are 1. For $b > t$ with m even, the denominator phases are 1. If T is even, the numerator phases are also 1. If T is odd, Proposition 9.28 gives $(\lambda')^2 = 1 + \lambda_0$, and (55) yields

$$H_{\chi_K} H_\tau = H(\lambda') H(\lambda_0) = H(1 + \lambda_0) H(\lambda_0) = 1.$$

Thus $R_H = 1$ in each of the three cases.

In these three cases, the trace-ideal formula (25) gives

$$w \in \mathfrak{p}_F^{\lceil m/2 \rceil} \cap \mathfrak{p}_F^{\lceil T/2 \rceil}.$$

Thus the stationary identities give

$$\chi_F(1 - w)^{-1} = \psi_F(An_u w), \quad \tau(1 + w)^{-1} = \psi_F(-Aw).$$

Now suppose that $b > t$ and m is odd. The τ -evaluation is still $\psi_F(-Aw)$, but the critical-function formula gives

$$\chi_F(1 - w)^{-1} = \psi_F(An_u w) h_\lambda(1).$$

The critical functions of χ_F and $\tau\chi_F$ are both h_λ , and (55) gives

$$R_H = H(\lambda)^{-2} = h_\lambda(1)^{-1}.$$

Hence R_H cancels the factor $h_\lambda(1)$.

In all the cases considered here, we therefore obtain

$$\mathcal{E}_{K/F}(\chi_F) = \psi_F(-As + An_u w - Aw) = \psi_F(-AX).$$

In equal characteristic two, $X = 0$ by (78). In mixed characteristic, the trace-ideal bounds from (25) give $X \in \mathfrak{p}_F^T$; see [4, Lemma 9.31]. Thus (75) gives $\mathcal{E}_{K/F}(\chi_F) = 1$.

The odd boundary. Suppose that $b = t$ and T is odd. This case occurs only in mixed characteristic, since T is even in equal characteristic two by [4, Lemma 9.30].

Here u is a unit. Put $\rho = \bar{u} \in k^\times$, so $\bar{n}_u = \rho^2$. Since χ_F has minimal conductor in its $S(K/F)$ -orbit, τ and $\tau\chi_F$ both have conductor T , so their stationary coefficients A and $A(1 + n_u)$ have the same valuation. Thus $1 + \rho^2 \neq 0$ and hence $1 + \rho \neq 0$. Put

$$B = \frac{\rho^2}{1 + \rho^2}, \quad C = \frac{\rho}{1 + \rho^2}.$$

The coefficients $\lambda_0, \lambda, \lambda_1, \lambda'$ satisfy (56) for this ρ , by [4, Proposition 9.28]. For $w = s/(1 + n_u)$, the evaluations in [4, proof of Proposition 9.29] are

$$\begin{aligned} \chi_F(1 - w)^{-1} &= \psi_F(An_u w) h_\lambda(B), \\ \tau(1 + w)^{-1} &= \psi_F(-Aw) h_{\lambda_0}(C), \\ \psi_F(-AX) &= \psi_0(C). \end{aligned}$$

Substituting these evaluations into (77) gives

$$(79) \quad \mathcal{E}_{K/F}(\chi_F) = R_H h_\lambda(B) h_{\lambda_0}(C) \psi_0(C).$$

Here R_H is the quotient R in (57). These coefficient relations allow us to apply (58) to (79), which gives $\mathcal{E}_{K/F}(\chi_F) = 1$.

The cancellation of R_H with the critical-function values outside the odd boundary is proved by `WildQuadraticCoefficientData.phaseRatio_mul_corrections_eq_one_of_not_boundaryOdd`. The complete nonboundary and boundary arguments are formalized by `wildQuadratic_nonboundary` and `wildQuadratic_boundary`, respectively. Together they prove `firstMain_wildQuadratic`.

6.7. Combining the cases. Sections 6.1–6.6 cover the following mutually exclusive possibilities:

extension and conductor	theorem applied
K/F unramified	unramified case
K/F tamely ramified, ℓ odd	tame odd case
K/F tamely ramified, $\ell = 2$	tame quadratic case
K/F wild, $m \leq 1$	endpoint cases
K/F wild, p odd, $2T \leq m$	stable case
K/F wild, p odd, $1 < m < 2T$	non-stable odd case
K/F wild, $p = 2$, $m > 1$	wild quadratic case.

The theorem `firstMainLemma_nonarchimedean` determines which case applies and invokes the corresponding result. Finally, `firstMainLemma` applies this result with `localConstantF` and `localConstantK`, and the conductors of the given characters, proving Theorem 1.1.

7. RESULTS OF POSSIBLE INDEPENDENT USE

The following formalized results may be useful elsewhere.

Result	Lean declaration
The finite Gauss sum occurring in Δ is nonzero, and the finite and Haar-integral realizations of Δ agree; see [4, Proposition 2.4].	<code>finiteGaussSum_ne_zero</code> <code>deltaIntegral_eq_deltaFinite</code>
For a ramified cyclic extension of prime degree, norm is an isomorphism on the graded unit groups away from the break, and $ F^\times/N_{K/F}(K^\times) = [K : F]$ by (6). See [4, Theorem 3.1 and Corollary 3.9].	<code>cyclicPrimeNormFiltration</code> <code>normQuotient_card</code>
At a positive wild break, the critical graded norm is $z \mapsto z^p - \lambda^{p-1}z$, as in (32). See [4, Proposition 3.3].	<code>criticalNormPolynomialValue_exact_of_positiveBreak</code>
The stationary class exists uniquely, and the right-hand side of Lamprecht's odd-conductor formula is independent of the chosen stationary representative; see [4, Lemma 4.2 and Lemma 4.6].	<code>stationaryNumeratorClass_existsUnique</code> <code>lamprechtOdd_complete_representative_independent</code>
For a Hasse function with nontrivial polar character and its lift $\phi_{k'/k}$, $-\sum_{x \in k'} \phi_{k'/k}(x) = (-\sum_{x \in k} \phi(x))^{[k':k]}$ by (44). See [4, Theorem 5.5].	<code>hasseFunctionLift</code>
The affine-pencil identity (51) holds under the hypotheses of Section 5.3.5; see [4, Lemma 5.7].	<code>affinePencil</code>

These declarations use the interfaces of the present project. A future contribution to Mathlib would require comparison with the existing Mathlib interfaces and, where appropriate, a more general statement.

8. ARTIFACT AND REPRODUCIBILITY

The formal artifact [5] is available as release **v1.0.0** of the GitHub repository github.com/fukubillueda-web/epsilon_FML, at commit

7a6cdfa23b5d246919b0d38f6953104756507a42.

It uses Lean v4.32.2 and Mathlib revision

905b95818eb32af7874a58b427f50c1711a5e96c.

The source is distributed under the Apache License 2.0.

With Git, elan/Lean, and Python 3 installed, a fresh checkout can be verified on macOS by running

```
git clone https://github.com/fukubillueda-web/epsilon_FML.git
cd epsilon_FML
git checkout v1.0.0
bash FIRST_RUN_MAC.command
```

The project introduces no axioms beyond the standard axioms used by Lean and Mathlib.

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